Sharp Aggregate Bounds for Cluster-Induced Split Leakage: A Public ciFAIR Case Study

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***Abstract*—When record-level cluster memberships cannot be shared, an auditor may disclose only population size M, number of non-singleton groups G, their total membership S, and fixed test size T. Known-cluster hypergeometric formulas do not identify leakage because individual group sizes remain hidden. This paper proves sharp closed-form extrema for expected crossing groups and exposed test files over every compatible integer allocation, plus a sharp refinement when a maximum group size may also be disclosed. The crossing result holds for any nontrivial split; the exposure result allows groups larger than the test set when the test side is strictly smaller than the training side. A separate exact dynamic program checks the extrema. A public ciFAIR illustration evaluates the calculations, illustrative certificate resolution, and component-merger sensitivity. For CIFAR-10, known-size expectations of 83.573 crossing groups and 86.097 exposed files lie inside aggregate-only bounds 80.722–84.446 and 85.390–86.668. CIFAR-100 gives 246.137 and 255.795 inside 231.559–248.615 and 252.226–257.504. The result is sharp partial identification of cluster-induced evaluation dependence, not a privacy guarantee, causal accuracy estimate, or duplicate detector.**

***Index Terms*—data leakage, duplicate clusters, partial identification, finite-population sampling, majorization**

I. Introduction

Train–test overlap can bias evaluation [1]. Separately, label errors can destabilize benchmark comparisons [2]. Duplicate audits have documented overlap and quantified its evaluation impact in CIFAR and face datasets [3], [4], while recent visual audits scale retrieval to large benchmarks [5], [6]. Group-aware splitting is preferable when memberships are available, but restricted datasets may prohibit their disclosure.

This paper asks: *What split leakage is identifiable when only (M, G, S, T) may be published?* Its contributions are: (1) sharp expectation extrema over every hidden group-size allocation, including an optional maximum-size disclosure refinement; (2) an exact allocation dynamic program and a regime-wide certificate-utility study; and (3) a reproducible implementation illustration on split-conditioned ciFAIR metadata, with annotation-policy, deletion, and component-merger sensitivity. The estimand is a counterfactual uniform fixed-size split, not classifier degradation.

II. Related Work

Classical finite-population richness work derives the expected number D_t of distinct classes observed in a size-t sample with arbitrary class frequencies [7], [8]; database selectivity work gives the same unequal-frequency expectation and an earlier equal-multiplicity specialization [9], [10]. A crossing count obeys the pointwise identity $C = D_T + D_{M-T} - G$, so the underlying expected-distinct functionals are not new. Guignard *et al.* derive multivariate-hypergeometric cluster allocations and the known-size exposed-element expectation and variance [11]; (2) below is its per-group expectation specialization. The new object is the integer fiber $\{(k_1, \dots, k_G) : k_i \geq 2, \sum k_i = S\}$ when the size vector is unavailable. The contribution is the partial-identification layer: discrete-concavity proofs and sharp, attainable extrema for two functionals over that fiber. Wong–Yue establish a related majorization order for distinct outcomes in independent repeated trials [12]; this is a conceptual antecedent, not the present fixed-size without-replacement model. Here crossing's Schur direction follows from the discrete-concavity proof below. The additional work here is to identify and attain the endpoint vectors over the disclosure fiber, derive the exposure endpoint, and handle an optional cap; Marshall supplies the general majorization machinery [13]. DataSAIL instead constructs splits from known similarities [14]. To the author's knowledge, this aggregate-only optimization has not appeared in the literature searched through Aug. 3, 2026; the archived search log records the expanded sampling, ecology, and database queries. This scoped claim asserts neither new sampling formulas nor a new majorization method. Table I makes the distinction explicit: Theorem 1 applies classical majorization to a new disclosure-limited information set and supplies attainable integer-fiber endpoints for both audit functionals.

III. Public Data and Operational Graphs

ciFAIR publishes CIFAR-10 and CIFAR-100 duplicate-pair annotations [3], [15]; CIFAR population and split sizes follow the primary dataset report [16]. At repository commit `4c2764277f5fda8fec6784a78c1818eab13236c5`, two CSVs per dataset list train–test and test–test pairs. Judgments denote approximate pixel copies (0), same-shot near-duplicates (1), and highly similar scenes (2).

Train and test identifiers use separate namespaces. The primary policy unions categories 0–2 and closes edges into

TABLE I

CLOSEST ANTECEDENTS AND THE PRECISE ADDED LAYER. "FULL VECTOR" MEANS INDIVIDUAL CLASS OR GROUP SIZES ARE KNOWN.

Antecedent	Functional	Information assumed	Result relative to this work
Hurlbert; Walton [7], [8]	Expected distinct classes	Full abundance vector	Supplies the known-vector sampling functional, not extrema over fixed (G, S)
Christodoulakis; Bitton–DeWitt [9], [10]	Database distinct-value expectation	Unequal frequencies or equal multiplicity	No sharp endpoints for every hidden integer allocation
Wong–Yue; Marshall [12], [13]	Related majorization tools	Independent repeated trials / abstract Schur order	Conceptual antecedents only; the fixed-size model and its feasible-fiber witnesses, exposure endpoint, and cap are supplied here
Guignard *et al.* [11]	Cluster leakage expectation/variance	Full cluster sizes	Known-size target; no (M, G, S, T) partial-identification interval

connected components. Components of size at least two are operational dependence groups; unlisted files are residual records. Closure produces the disjoint groups required by the theorem, but similarity itself is not transitive and the components are not claimed as identities or ground-truth provenance classes. No image download or feature extraction is used. Both populations have $M = 60{,}000$ and $T = 10{,}000$ (Table II).

IV. SHARP AGGREGATE-ONLY BOUNDS

Let M, T, G, S be integers with $1 \leq T < M$, $G \geq 1$, and $2G \leq S \leq M$. Hidden group sizes $k_i \geq 2$ satisfy $\sum_i k_i = S$. Under a uniformly sampled test set of size T, a size-k group crosses train and test with probability

$$c(k) = 1 - \frac{\binom{M-k}{T}}{\binom{M}{T}} - \frac{\binom{M-k}{T-k}}{\binom{M}{T}}, \tag{1}$$

and its expected test files having a same-group training record are

$$e(k) = \frac{kT}{M} - k\frac{\binom{M-k}{T-k}}{\binom{M}{T}}. \tag{2}$$

Out-of-range binomial coefficients are zero. Let $K = S - 2(G-1)$ be the largest feasible group.

Theorem 1. (a) For every $1 \leq T < M$, $c(k)$ is nondecreasing and discrete-concave on $2 \leq k \leq K$. (b) If $T \leq (M-1)/2$, $e(k)$ is strictly increasing and discrete-concave on the same domain; no $K \leq T$ condition is needed. Consequently, the stated extrema hold for $\sum_i c(k_i)$ always and for $\sum_i e(k_i)$ under condition (b): each is minimized by $(K, 2, \ldots, 2)$ and maximized by group sizes differing by at most one.

Corollary 1 (optional size cap). If an auditor additionally discloses $2 \leq k_i \leq H$ with $S \leq GH$ and $H > 2$, write $E = S - 2G = q(H-2) + r$, $0 \leq r < H - 2$. The balanced allocation remains the sharp maximum, while the sharp minimum has q groups of size H, one of size $2 + r$ when $r > 0$, and all others of size 2. (For $H = 2$, feasibility forces all groups to size 2.) This follows by the same reverse exchanges, stopped at the cap. *Proof of Theorem 1 and Corollary 1.* Write $(q)_k = q(q-1)\cdots(q-k+1)$, extended as zero for $k > q$, and define $a_k = (T)_k/(M)_k$, $b_k = (M-T)_k/(M)_k$, and $L = M - T$. For $u_q(k) = (q)_k/(M)_k$ and $2 \leq k \leq K-2$ (when this range is nonempty), direct subtraction gives

$$\Delta^2 u_q(k) = u_q(k)\frac{(M-q)(M-q-1)}{(M-k)(M-k-1)} \geq 0, \tag{3}$$

including the boundary by zero extension. Hence $c(k) = 1 - a_k - b_k$ is concave. Adding a fixed group member under a coupled split cannot destroy a crossing, so c is nondecreasing.

Define $\Delta f(k) = f(k+1) - f(k)$ and $\Delta^2 f(k) = f(k+2) - 2f(k+1) + f(k)$; if $K < 4$, the second-difference condition is vacuous. The displayed formula remains valid at $k = q-1$ and $k = q$ because $u_q(q+1) = u_q(q+2) = 0$, and it is zero after the truncation. For exposure and $2 \leq k \leq \min\{T, K-2\}$,

$$\Delta^2(ka_k) = a_k\frac{L[k(L+1) - 2T]}{(M-k)(M-k-1)} > 0, \tag{4}$$

because $L \geq T + 1$ and $k \geq 2$. At $k = T - 1$ and $k = T$, zero extension gives the same positive expression; for $k > T$, $ka_k = 0$. Thus $e(k) = kT/M - ka_k$ is concave. Its differences equal $T/M > 0$ beyond T and, by concavity, are no smaller before T, proving strict increase, including $T = 1$. Finally, for concave f and $x \geq y + 2$, balancing changes the sum by $\Delta f(y) - \Delta f(x-1) \geq 0$. Repetition yields the balanced maximum; reverse exchanges yield the concentrated minimum [13]. These allocations attain the endpoints; uniqueness is not claimed because either functional can have affine or constant tails.

A separate exact check sets $E = S - 2G$, $D_0(0) = 0$, and

$$D_g(n) = \operatorname*{ext}_{0 \leq r \leq n}\{D_{g-1}(n-r) + f(2+r)\}, \tag{5}$$

TABLE II

PINNED CIFAIR CONSTRUCTION AND OFFICIAL-SPLIT OBSERVATION. HISTOGRAMS USE SIZE:COUNT.

Dataset	Edges	G	S	Max.	Group-size histogram	Crossing	Exposed
CIFAR-10	325	288	608	8	2:272, 3:8, 4:5, 6:2, 8:1	268	289
CIFAR-100	995	831	1,790	8	2:752, 3:51, 4:17, 5:4, 6:5, 7:1, 8:1	801	899

TABLE III

SHARP CROSSING CERTIFICATES AT $M = 60{,}000$, $G = 100$, $T/M = 1/6$. BOUNDS ARE PERCENTAGES OF G; WIDTH IS RELATIVE TO THE UPPER ENDPOINT.

Mean size	H	Sharp bounds	Width	Decision
3	–	[28.5, 41.7]	31.6%	low
5	–	[28.5, 59.8]	52.3%	inconclusive
5	8	[52.3, 59.8]	12.6%	high
10	16	[66.1, 83.9]	21.2%	high

with separate minima and maxima. This $O(GE^2)$ recurrence enumerates every excess-membership composition rather than relying on the closed-form witness. The artifact executes it for every nonempty public configuration and compares it with full enumeration on more than 500 small fibers.

V. WHEN ARE FOUR AGGREGATES DECISION-USEFUL?

A deterministic stress grid fixes $M = 60{,}000$ and $G = 100$, crosses test fractions 5%, 16.7%, and 33.3% with mean group sizes 2, 2.2, 3, 5, 10, and 20, and reports interval width normalized by its upper endpoint. Across the uncapped grid, crossing width reaches 83.8%, whereas exposure width reaches 12.3%. Four aggregates can therefore be decisive or weak depending on the regime; narrow ciFAIR intervals are not universal.

For an operational illustration, declare "high crossing risk" only when the lower bound is at least $0.5G$, "low" only when the upper bound is below $0.5G$, and otherwise return "inconclusive." Table III shows a concrete disclosure decision at $T/M = 1/6$: mean size 5 is inconclusive from four aggregates, but authorized disclosure of $H = 8$ raises the sharp lower endpoint above $0.5G$ and certifies high risk. The $0.5G$ threshold is illustrative, not policy-derived; this synthetic study demonstrates certificate mechanics, not validated operational utility or prevalence.

VI. IMPLEMENTATION ILLUSTRATION: SPLIT-CONDITIONED CIFAIR GRAPHS

The public metadata ascertain train–test and test–test relations but contain no train–train census; the operational graph is therefore conditioned on the official split. This section validates implementation and estimand sensitivity on that incomplete graph, not population duplicate prevalence. Table IV compares known-size expectations with bounds using only (M, G, S, T). Containment is a theorem consequence, so this is deterministic implementation validation, not an empirical hypothesis test. Exact pairwise inclusion–exclusion additionally gives crossing-count standard deviations of 7.617 and 12.915 for the known public histograms; variance is not claimed identifiable from the four aggregates. The crossing intervals span 4.46% and 6.93% of the known expectations, while exposure intervals span 1.48% and 2.06%.

The four-aggregate intervals are the least informative release tier. If maximum size $H = 8$ is also disclosed, Corollary 1 raises the crossing lower bound to 82.689 for CIFAR-10 and 241.359 for CIFAR-100, and the exposure lower bound to 85.665 and 253.389; balanced upper bounds are unchanged. This supplies a simple disclosure ladder: publish (M, G, S, T) first, add H only if its release is authorized and the initial interval is too wide, and otherwise retain the wider certified interval. The allocation interval, crossing SD, and merger stress represent distinct deterministic partial-identification, split-sampling, and graph-annotation uncertainties; none substitutes for another. Table V shows that annotation policy dominates hidden-size uncertainty: broadening the operational relation changes CIFAR-100 known crossing from 11.389 to 246.137. Category 0 has no CIFAR-10 edges; the displayed $[0, 0]$ interval is an explicit $G = S = 0$ convention outside Theorem 1. The result is therefore conditional on the declared graph policy, not robust to an unknown true duplicate relation. As a local one-edge perturbation diagnostic, deleting each published primary-policy edge in turn changes either known expectation by at most 0.278 for CIFAR-10 and 0.278 for CIFAR-100. This does not establish robustness to missing or systematically misclassified relations.

Proposition 2 (missing-edge taxonomy). Merging observed components of sizes $a, b \geq 2$ gives $\delta_c = c(a+b) - c(a) - c(b) \leq 0$ and $\delta_e = e(a+b) - e(a) - e(b) \geq 0$. In contrast, attaching a residual singleton to a size-a component changes the metrics by $c(a+1) - c(a) \geq 0$ and $e(a+1) - e(a) \geq 0$; linking two residual singletons creates $c(2), e(2) > 0$. Thus graph incompleteness has no single conservative direction. For disjoint components A, B, $\delta_c = \Pr(O) - \Pr(H)$, where O is the event that A, B are opposite-pure and H the event that both cross. Fix $x \in A, y \in B$: swapping their split assignments maps O injectively into H, preserves test size, and is inverted by the same swap, proving the first sign. The remaining signs follow pointwise: adding relations cannot destroy a crossing or remove a training partner, while merging two already-counted components can reduce their count.

Table VI quantifies one bounded existing-component-merger scenario under budgets 1, 10, and 25. Separate greedy paths choose the largest crossing decrease or exposure increase at each step; every selected size pair is frozen in the artifact. These are not joint cells, sharp missing-edge bounds, or

TABLE IV
KNOWN-SIZE VALUES VERSUS SHARP AGGREGATE-ONLY BOUNDS. PARENTHESES GIVE KNOWN-SIZE CROSSING SD.

Dataset	Known crossing (SD)	Crossing bounds	Known exposure	Exposure bounds
CIFAR-10	83.573 (7.617)	80.722–84.446	86.097	85.390–86.668
CIFAR-100	246.137 (12.915)	231.559–248.615	255.795	252.226–257.504

TABLE V
POLICY SENSITIVITY. EACH METRIC IS KNOWN VALUE / [AGGREGATE MINIMUM, MAXIMUM].

Dataset	Judgments	G	S	Crossing	Exposure
CIFAR-10	$\{0\}$	0	0	0.000/[0.000,0.000]	0.000/[0.000,0.000]
CIFAR-10	$\{0,1\}$	249	523	72.020/[69.883,72.640]	73.950/[73.390,74.376]
CIFAR-10	$\{0,1,2\}$	288	608	83.573/[80.722,84.446]	86.097/[85.390,86.668]
CIFAR-100	$\{0\}$	41	82	11.389/[11.389,11.389]	11.389/[11.389,11.389]
CIFAR-100	$\{0,1\}$	596	1,267	174.374/[166.281,175.975]	180.115/[178.114,181.183]
CIFAR-100	$\{0,1,2\}$	831	1,790	246.137/[231.559,248.615]	255.795/[252.226,257.504]

TABLE VI
EXISTING-COMPONENT MERGER STRESS. METRIC ROWS ARE DIFFERENT GREEDY TRAJECTORIES.

Dataset	Metric	1	10	25
CIFAR-10	Cross. drop	0.510	4.933	9.794
	Expos. rise	0.108	1.080	2.700
CIFAR-100	Cross. drop	0.553	6.206	13.961
	Expos. rise	0.108	1.080	2.700

merger-likelihood models.

The official splits contain 268/801 crossing components and 289/899 exposed test files. Because the annotations and official split are coupled, these observations are descriptive and are not causal comparisons with the random-split model.

VII. REPRODUCIBILITY, RIGHTS, AND LIMITATIONS

The tagged public artifact is available at https://github.com/N-Mahesh/MOSAIC-EZ/tree/urtc2026-cifair-v15. It contains the PDF, source, frozen claim JSON, generated cell macros, manifest, code, search log, and tests. The verified environment is Python 3.12.5 with pypdf 6.14.2; the full check exits nonzero on checksum, schema, claim, PDF, manifest, source-rebuild, or test failure and took 135 seconds on the source workstation. Table VII gives the tested command and exact upstream paths/hashes.

The pinned ciFAIR repository exposes no explicit license file for redistributing its metadata. The artifact therefore redistributes neither those CSVs nor CIFAR images; it stores hashes and derived aggregates and requires retrieval from the cited upstream source. Users remain responsible for upstream terms. No identifiable or new human-subject data are included.

The annotation graph omits a train–train census and is retrieval-biased. Proposition 2 shows why no generic conservative direction exists: merging observed components lowers crossing count, whereas attaching or linking residual singletons does not decrease it; exposure does not decrease in each case. The table covers only declared existing-component mergers. Missing nodes, false-positive edges, and the unknown ascertainment mechanism remain unbounded. Category 2 and transitive closure are operational choices. Aggregate-only expectations do not identify variance, tail risk, classifier-score inflation, or realized leakage. Publishing (M, G, S, T) is disclosure-limited but provides no formal privacy guarantee.

VIII. CONCLUSION

Discrete concavity yields sharp expectation intervals from (M, G, S, T) without memberships and without requiring the maximum-group-size condition $K \leq T$. The public case study shows narrow allocation uncertainty but large annotation-policy sensitivity. These bounds can support a disclosure-limited audit when memberships cannot be published, but group-aware splitting remains preferable and no causal accuracy or privacy claim follows.

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TABLE VII

ARTIFACT COMMAND AND PINNED UPSTREAM CIFAIR METADATA.

Command: `python tools/verify_public_release_bundle.py . --cifair-meta cifair/meta --rebuild-pdf`	
`meta/duplicates_cifar10.csv`	`4cb7e99a7dfff346082c9d8fa4c2989a 196e4a37d1e58f75936046696f1ba6a4`
`meta/duplicates_cifar10_test.csv`	`ef7b33e2f32d056cdb342046b437ee81 9541a07cb3bb0520a3ada9b9c035f532`
`meta/duplicates_cifar100.csv`	`3891498a862f2d73df00cd5dc2ee9ae2 b6dccf5238691d3c29d9abb12e1c63cb`
`meta/duplicates_cifar100_test.csv`	`78c9d8d17eda11b468a137119ed51f09 332393d65a6329035e2250888edb2760`

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